Uttapreksha Patel

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Research engineer who originates methodology and ships the code that proves it out: built CUDA kernels reaching 0.91–0.95× of official FlashAttention-2, found and fixed real bugs in my own eval harness, and implemented RL post-training from scratch. Published research, production-quality Python/CUDA daily.

Technical Skills

Research	Experimental Design, Statistical Analysis, Reproducibility
RL	PyTorch, GRPO/PPO/DPO (from scratch), Reward Design
Evals	Coding Benchmarks, LLM-as-Judge, Statistical QA, Judge Bias
Systems	CUDA C++, Docker, Kubernetes, Nsight Compute
Languages	Python (OOP, PEP8), C++ (pybind11), CUDA, SQL

Work Experience

Northeastern University | *Collaboration with Sandia National Laboratories* — **Boston, MA**
Research Assistant | Systems Engineering & Reproducible Analysis — *February 2025 – Present*

- Scoped and shipped Python data pipelines end-to-end without a fully specified brief, processing **1M+ operation records** with automated anomaly detection; deployed to production on AWS EC2, improving pipeline reliability **30%**
- Established whether a result holds before trusting it: designed controlled experiments isolating exactly which resource caused a performance regression, quantifying effects with sub-second precision across **8 identified events**
- Deployed telemetry infrastructure and diagnosed critical buffer-overflow failures through memory allocation tuning, restoring stable service under multi-node production load

TecUnique Pvt. Ltd. — **Vadodara, India**
Python Development Intern — *Dec 2023 – Apr 2024*

- Built Python REST API automation framework using Pytest achieving **93% pass rate across 30 test cases**; automated end-to-end validation with structured HTML reports and regression tracking
- Optimized SQL queries and numerical pipelines, reducing runtime **20%**; managed source control through Git in an Agile/Scrum environment

Relevant Projects

nanoserve: LLM Inference Kernels in CUDA C++, Systems Research | *CUDA C++, PyTorch, Nsight Compute, SASS | GitHub*

- Built an LLM inference stack from scratch in CUDA C++ (GEMM, FlashAttention-2, fused RMSNorm/RoPE/SwiGLU, 4-bit quantized GEMM, Mixture-of-Experts), integrated into PyTorch to run Qwen2.5-1.5B and 7B
- Achieved **2.61× faster** token generation than Hugging Face Transformers on Qwen2.5-7B; implemented a FlashAttention-2 kernel reaching **0.91–0.95×** of the official implementation and a tensor-core GEMM at **90% of cuBLAS**, diagnosing each bottleneck with Nsight Compute and SASS inspection across 14 documented iterations

Secure Code Execution Sandbox for Coding Evals: Benchmark Architecture, Self-Audited | *Python, Docker, Linux Process Isolation, pass@k Evaluation | GitHub*

- Originated and shipped an evaluation architecture: a Docker sandbox for untrusted model-generated code with an out-of-process test oracle, closing exploits like early `exit(0)` and always-true `__eq__` that fool exit-code harnesses; evaluated Qwen2.5-Coder-1.5B on 3,280 HumanEval samples
- Found and fixed three bugs in my own harness through the study, each mislabeling failures; same-seed regeneration confirmed **87%** of flagged errors were genuine model mistakes, not artifacts of my own test setup, establishing that the result actually held

RLHF Preference Data QA Pipeline: Reproducible Evaluation, Judge Calibration | *Python, LLM-as-Judge, Statistical QA, HH-RLHF | GitHub*

- Built an automated QA pipeline for HH-RLHF preference data combining embedding-based heuristics with a local LLM judge (Qwen2.5-7B); injected 5 types of known-bad examples to measure detection, raising label-flip recall from 10% to 57% and truncation recall from 29% to 88% over heuristics alone
- Measured **32.7% position bias** in the LLM judge across 300 pairs; simulated 12 weeks of annotation campaigns with statistical process control, catching a quality-drop the same week it occurred

GRPO-LLM: RL Methods, Built Entirely from Scratch | *Python, PyTorch, C++/pybind11, Kubernetes, AWS | GitHub | Blog*

- Implemented GRPO from scratch without wrapping TRL; ran a controlled 3-way comparison against PPO and DPO, isolating reward sparsity, not compute, as the actual bottleneck; profiled and validated a C++ optimization measuring a **1.7× speedup** with **25 passing tests**

Research

Analyzing VLM-Based Approaches for Anomaly Classification and Segmentation | *Co-authored, arXiv 2026*

- Evaluated WinCLIP and AnomalyCLIP on MVTec AD and VisA benchmarks; achieved **0.916 classification AUROC** and **0.907 segmentation AUROC** through reproducible evaluation pipelines and systematic error analysis

Education

Northeastern University — **Boston, MA**
M.S. Electrical and Computer Engineering (GPA: 3.9/4.0) — *May 2026*

G H Patel College of Engineering and Technology — **Anand, India**
B.E. Electronics and Communication Engineering (GPA: 3.7/4.0) — *May 2024*